

Partial Derivatives

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1 Introduction

Before understanding partial derivatives, let's recall and develop our understanding of derivatives of a single variable. Before continuing, I invite the reader to imagine what a derivative is.

If you thought something similar to a tool for linear approximation, or the slope of the tangent line, or even velocity, or acceleration, these are all good thoughts. However, let's focus on the frame of linear approximation. When we zoom in around a point on any differentiable function, the function becomes similar to a line. This is represented by the derivative evaluated at that point where the derivative is the slope.

Now, we generalize from two dimensions into three dimensions. So our approximation of a two dimensional mapping having a one dimensional line representing the derivative generalizes into a two dimensional plane when considering a three dimensional mapping. Why is this? Let's consider an implication regarding derivatives of one variable being a method for linearly approximation to the given function $f : \mathbb{R} \rightarrow \mathbb{R}$. This means, that by applying a small enough bound on an interval, we find any differentiable function $f' : \mathbb{R} \rightarrow \mathbb{R}$. The function f' having one input and output approximates locally one dimensional space. Therefore, to generalize to higher dimensions, we expect our derivatives to represent a space of the same dimension as the number of inputs to the given function. Then, this implies looking at a function of two variables $g : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ with $g(x, y)$, would exist locally in approximately two dimensional euclidean space.

Then, this would imply that, as we generalize to higher dimensions, we would expect the derivative to be representative of a tangent space that has the same dimension as the number of inputs to the function. This concept is the idea of a gradient and will be developed more in the gradient review.

2 Definitions and Theorems

Definition of Continuous

Let $A \subseteq \mathbb{R}$ and $f : A \rightarrow \mathbb{R}$ be a function. We call f continuous at a point $x_0 \in A$, if and only if, the following hold:

$$\begin{aligned}f(x_0) &= y \\ \lim_{x \rightarrow x_0} f(x) &= c \\ c &= y\end{aligned}$$

for some $y, c \in \mathbb{R}$. Furthermore, f is continuous on A if f is continuous at every $x_0 \in A$.

(For a more formal definition, look at the calculus 1 and 2 review or the Wikipedia page for continuous functions)

Definition of Differentiable

Let $A \subseteq \mathbb{R}$ and $f : A \rightarrow \mathbb{R}$ be a function. We say f , is differentiable at a point $x_0 \in A$ if the limit, L , exists

$$L = \lim_{x \rightarrow x_0} \frac{f(x+h) - f(x)}{h} = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0}$$

If f is differentiable $\forall x_0 \in A$ the f is differentiable on A .

(For a more formal definition, look at the Wikipedia page for differentiable functions)

Definition of the Partial Derivative

Let $A \subseteq \mathbb{R}^n$ and $f : \mathbb{R}^n \rightarrow \mathbb{R}$. Then the partial derivative with respect to the i^{th} coordinate

$$f_{x_i} = \frac{\partial f}{\partial x_i} := \lim_{h \rightarrow 0} \frac{f(x_1, x_2, \dots, x_i + h, \dots, x_n) - f(x_1, x_2, \dots, x_n)}{h}$$

Definition of the Directional Derivative

Let $A \subseteq \mathbb{R}$ and $f : A^n \rightarrow \mathbb{R}$ be a differentiable function. Suppose $\vec{u} = \begin{bmatrix} u_1 \\ \dots \\ u_n \end{bmatrix}$ be a vector. Then we define the directional derivative

$$D_{\vec{u}} f(x_1, \dots, x_n) := \lim_{h \rightarrow 0} \frac{f(x_1 + u_1 h, \dots, x_n + u_n h) - f(x_1, \dots, x_n)}{h}$$

Additionally, we will define implicit and explicit equations. This property informs how we will take the partial derivative.

Definition of Explicit Equations

Consider a relation between two variables: x and y . Say x is independent and y is dependent on x , then $y = f(x)$. We define an equation of this form to be an explicit equation.

An example of an explicit equation is $y = \cos^2(x) + 4$

Definition of Implicit Equations

Consider a relation between two variables x and y . If the relation can not be solved, is not solved for x in terms of y , or visa versa, we call this an implicit equations and is of the form $f(x, y) = 0$.

An example of an implicit equation is $\sin(x) + xy - \pi = 0$

Theorem Chain Rule

Let $A \subseteq \mathbb{R}$ and $g_0, \dots, g_n : A \rightarrow \mathbb{R}$ and $f : A^n \rightarrow \mathbb{R}$. Suppose $z = f(x_0, \dots, x_n)$ and $x_0 = g_0(t), \dots, x_n = g_n(t)$ be differentiable functions. Then $z = f(g_0, \dots, g_n)$ is differentiable with respect to t and

$$\frac{dz}{dt} := \sum_{i=0}^n \frac{\partial z}{\partial x_i} \frac{dx_i}{dt}$$

where the ordinary derivatives are evaluated at t and the partial derivatives are evaluated at $(x_0, \dots, x_n)$.

Definition of the Gradient

Let $A \subseteq \mathbb{R}$ and $f : A^n \rightarrow \mathbb{R}$ be a function such that $f_{x_1}, \dots, f_{x_n}$ exists. (recalling the notation for partial derivatives) The vector $\nabla f(x_1, \dots, x_n)$ is called the gradient of f and is defined as

$$\nabla f(x_1, \dots, x_n) := \begin{bmatrix} f_{x_1}(x_1, \dots, x_n) \\ \dots \\ f_{x_n}(x_1, \dots, x_n) \end{bmatrix}$$

3 Taking a Partial Derivative

When taking a partial derivative we hold all variables constant except the variable with which we are differentiating with respect to.

The first example we will look at is the unit circle, $x^2 + y^2 = 1$. Observe this equation cannot be put into explicit form nicely as we would lose half the circle due to the square root. For more depth on this read the next paragraph. Otherwise skip if you already know why there is a plus minus on $\sqrt{a^2}$ but not on $\sqrt{a}$.

(This is due to the domain restriction on the square root function. Consider $f(x) = \sqrt{x}$ and $g(x) = x^2$. The we know $g(x)$ has domain $(-\infty, \infty)$ and range (often called co-domain) $[0, \infty)$. This is written as $g : \mathbb{R} \rightarrow \mathbb{R}^+$ where $\mathbb{R}^+ = \mathbb{R}_{\geq 0} = \{x \in \mathbb{R} | x \geq 0\}$. Similarly, $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ meaning the square root function has domain and co-domain of the non negative real numbers. Therefore $g \circ f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ as the co-domain of f must be a subset of the domain of g , which is true as $\mathbb{R}^+ \subset \mathbb{R}$ therefore for composition of functions we take the domain of the inside function and the co-domain of the outside function. For this reason when we switch the order and look at $f \circ g : \mathbb{R} \rightarrow \mathbb{R}^+$ we again have the case where the co-domain of g is a subset of the domain of all real numbers $\mathbb{R}$ and a co-domain of non negative real numbers $\mathbb{R}^+$. So given that f and g being the square and square root functions are inverses of each other we need to account for every element of the domain of $f \circ g$ which leads to $(f \circ g)(x) = \sqrt{x^2} = |x|$ thus when dropping the absolute value or square square root functions we are left with a $\pm$ on our result giving $(f \circ g)(x) = \pm x$.)

Returning to the unit circle $x^2 + y^2 = 1$ cannot be written in explicit form and is not yet in proper implicit form so lets correct that and write $x^2 + y^2 - 1 = 0$. Let's now take the partial derivative with respect to x .

$$0 = x^2 + y^2 - 1$$

Now that we are in implicit form we can write an expression instead of an equation.

$$f(x, y) = x^2 + y^2 - 1$$

$$\frac{\partial}{\partial x} f(x, y) = \frac{\partial}{\partial x} (x^2 + y^2 - 1)$$

Remember y is constant so,

$$= \frac{\partial}{\partial x} (x^2) + \frac{\partial}{\partial x} (y^2 - 1)$$

$$= \frac{\partial}{\partial x} (x^2) + 0$$

$$= \frac{\partial}{\partial x} (x^2)$$

$$f_x(x, y) = 2x$$

Similarly, $f_y(x, y) = 2y$. So if we now look at the gradient $\nabla f(x, y)$ we have:

$$\nabla f(x, y) = \begin{bmatrix} f_x(x, y) \\ f_y(x, y) \end{bmatrix}$$

Which means our gradient is:

$$\nabla f(x, y) = \begin{bmatrix} 2x \\ 2y \end{bmatrix}$$

Lets now look at the directional derivative through the lens of the definition.

$$D_{\vec{u}}f(x_1, \dots, x_n) = \lim_{h \rightarrow 0} \frac{f(x_1 + u_1h, \dots, x_n + u_nh) - f(x_1, \dots, x_n)}{h}$$

for our purposes lets consider the directional given by the vector $\begin{bmatrix} 3 \\ 4 \end{bmatrix}$. Although the definition does not restrict us to a unit vector for the purpose of engineering we usually don't want to scale our derivatives by some magnitude associated with our direction of interest so we will unitize the direction given by the vector. Our chosen vector has unit $\begin{bmatrix} \frac{3}{5} \\ \frac{4}{5} \end{bmatrix}$. So

$$\begin{aligned} D_{\vec{u}}f(x, y) &= \lim_{h \rightarrow 0} \frac{f(x + \frac{3}{5}h, y + \frac{4}{5}h) - f(x, y)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(x + \frac{3}{5}h)^2 + (y + \frac{4}{5}h)^2 - 1 - (x^2 + y^2 - 1)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(x + \frac{3}{5}h)^2 + (y + \frac{4}{5}h)^2 - 1 - x^2 - y^2 + 1}{h} \\ &= \lim_{h \rightarrow 0} \frac{(x + \frac{3}{5}h)^2 + (y + \frac{4}{5}h)^2 - x^2 - y^2}{h} \end{aligned}$$

$$\begin{aligned}
&= \lim_{h \rightarrow 0} \frac{(x + \frac{3}{5}h)^2 + (y + \frac{4}{5}h)^2 - x^2 - y^2}{h} \\
&= \lim_{h \rightarrow 0} \frac{x^2 + 2x\frac{3}{5}h + (\frac{3}{5}h)^2 + y^2 + 2y\frac{4}{5}h + (\frac{4}{5}h)^2 - x^2 - y^2}{h} \\
&= \lim_{h \rightarrow 0} \frac{2x\frac{3}{5}h + (\frac{3}{5}h)^2 + 2y\frac{4}{5}h + (\frac{4}{5}h)^2}{h} \\
&= \lim_{h \rightarrow 0} \frac{\frac{6}{5}xh + \frac{9}{25}h^2 + \frac{8}{5}yh + \frac{16}{25}h^2}{h} \\
&= \lim_{h \rightarrow 0} \frac{h(\frac{6}{5}x + \frac{9}{25}h + \frac{8}{5}y + \frac{16}{25}h)}{h} \\
&= \lim_{h \rightarrow 0} \frac{6}{5}x + \frac{9}{25}h + \frac{8}{5}y + \frac{16}{25}h
\end{aligned}$$

Which leaves the directional derivative

$$D_{\vec{u}}f(x, y) = \frac{6}{5}x + \frac{8}{5}y$$

We could also get the directional derivative using the chain rule by plugging in the parameter $x(t) = \frac{3}{5}t$ and $y(t) = \frac{4}{5}t$, as we are approaching along a line that is of the direction of the directional derivative. This is because we have parameterized a line that is equivalent to our direction. So applying the chain rule

$$\frac{dz}{dt} = \sum_{i=0}^n \frac{\partial z}{\partial x_i} \frac{dx_i}{dt}$$

we have

$$\begin{aligned}
\frac{df}{dt} &= \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} \\
&= \frac{\partial}{\partial x}(x^2 + y^2 - 1) \frac{d}{dt}(\frac{3}{5}t) + \frac{\partial}{\partial y}(x^2 + y^2 - 1) \frac{d}{dt}(\frac{4}{5}t)
\end{aligned}$$

$$= 2x\frac{3}{5}t + 2y\frac{4}{5}t$$

So we have from the chain rule

$$\frac{df}{dt} = \frac{6}{5}x + \frac{8}{5}y$$

and since we used a line parametrized from the chosen direction of our directional derivative example

$$\frac{df}{dt} = D_{\vec{u}}f(x, y) = \frac{6}{5}x + \frac{8}{5}y$$

Let's now look at one more example which is slightly more complicated and related to fluid mechanics. Consider the θ - β -Mach relation given as:

$$\tan(\theta) = 2\cot(\beta) \frac{M^2 \sin^2(\beta) - 1}{M^2[\gamma + \cos(2\beta)] + 2}$$

Where θ is the angle between the object and the oncoming flow, β is the angle between the weak shock and the oncoming flow and M is the mach number taken at the oncoming flow.

Let's first make note of the form of the θ - β -Mach relation. It is not in explicit or implicit form. Looking we will observe that by solving for $\theta(\beta, M)$ as a function of β and M we do not lose any information meaning we can safely write the θ - β -Mach relation in explicit form as

$$\theta = \arctan[2\cot(\beta) \frac{M^2 \sin^2(\beta) - 1}{M^2[\gamma + \cos(2\beta)] + 2}]$$

Starting with the partial with respect to M .

$$\frac{\partial \theta}{\partial M} = \frac{\partial}{\partial M} \arctan[2\cot(\beta) \frac{M^2 \sin^2(\beta) - 1}{M^2[\gamma + \cos(2\beta)] + 2}]$$

let

$$u = 2\cot(\beta) \frac{M^2 \sin^2(\beta) - 1}{M^2[\gamma + \cos(2\beta)] + 2}$$

and recall

$$\frac{d}{du} \tan(u(t)) = \frac{1}{[u(t)]^2 + 1} u'(t)$$

so we have

$$\frac{\partial \theta}{\partial M} = \frac{1}{[2\cot(\beta) \frac{M^2 \sin^2(\beta) - 1}{M^2[\gamma + \cos(2\beta)] + 2}]^2 + 1} \frac{\partial}{\partial M} [2\cot(\beta) \frac{M^2 \sin^2(\beta) - 1}{M^2[\gamma + \cos(2\beta)] + 2}]$$

Now recall the quotient rule

$$\frac{d}{du} \frac{f(u)}{g(u)} = \frac{f(u)g'(u) - f'(u)g(u)}{g(u)^2} u'$$

so we have

$$\frac{\partial \theta}{\partial M} = \frac{1}{[2\cot(\beta) \frac{M^2 \sin^2(\beta) - 1}{M^2[\gamma + \cos(2\beta)] + 2}]^2 + 1} 2\cot(\beta) \frac{(M^2 \sin^2(\beta) - 1)(2M[\gamma + \cos(2\beta)]) - 2M \sin^2(\beta)(M^2[\gamma + \cos(2\beta)] + 2)}{(M^2[\gamma + \cos(2\beta)] + 2)^2}$$

As this partial does not simplify nicely, we will end our work here. Our next partial $\frac{\partial \theta}{\partial \beta}$ is more useful for engineering as when this is equal to zero a shock wave transitions from normal to strong. Now taking the partial derivative with respect to β .

$$\begin{aligned} \frac{\partial \theta}{\partial \beta} &= \frac{\partial}{\partial \beta} \arctan[2\cot(\beta) \frac{M^2 \sin^2(\beta) - 1}{M^2[\gamma + \cos(2\beta)] + 2}] \\ &= \frac{1}{[2\cot(\beta) \frac{M^2 \sin^2(\beta) - 1}{M^2[\gamma + \cos(2\beta)] + 2}]^2 + 1} \frac{\partial}{\partial \beta} [2\cot(\beta) \frac{M^2 \sin^2(\beta) - 1}{M^2[\gamma + \cos(2\beta)] + 2}] \\ &= \frac{1}{[2\cot(\beta) \frac{M^2 \sin^2(\beta) - 1}{M^2[\gamma + \cos(2\beta)] + 2}]^2 + 1} \left[\frac{2\cot(\beta)(M^2 \sin^2(\beta) - 1)}{M^2[\gamma + \cos(2\beta)] + 2} \right] \end{aligned}$$

recall the derivative

$$\frac{d}{dx} \cot(x) = -\sec^2(x)$$

and

$$\frac{d}{dx} \sin^2(x) = 2\sin(x)\cos(x)$$

so the partial of θ with respect to β is as follows

$$= \frac{(M^2[\gamma + \cos(2\beta)] + 2)(-2\sec^2(\beta)(M^2 \sin^2(\beta) - 1) + 2\cot(\beta)M^2 2\sin(\beta)\cos(\beta)) - 2\cot(\beta)(M^2 \sin^2(\beta) - 1)(-2M^2 \sin(2\beta))}{([2\cot(\beta) \frac{M^2 \sin^2(\beta) - 1}{M^2[\gamma + \cos(2\beta)] + 2}]^2 + 1)(M^2[\gamma + \cos(2\beta)] + 2)^2}$$